



PennState

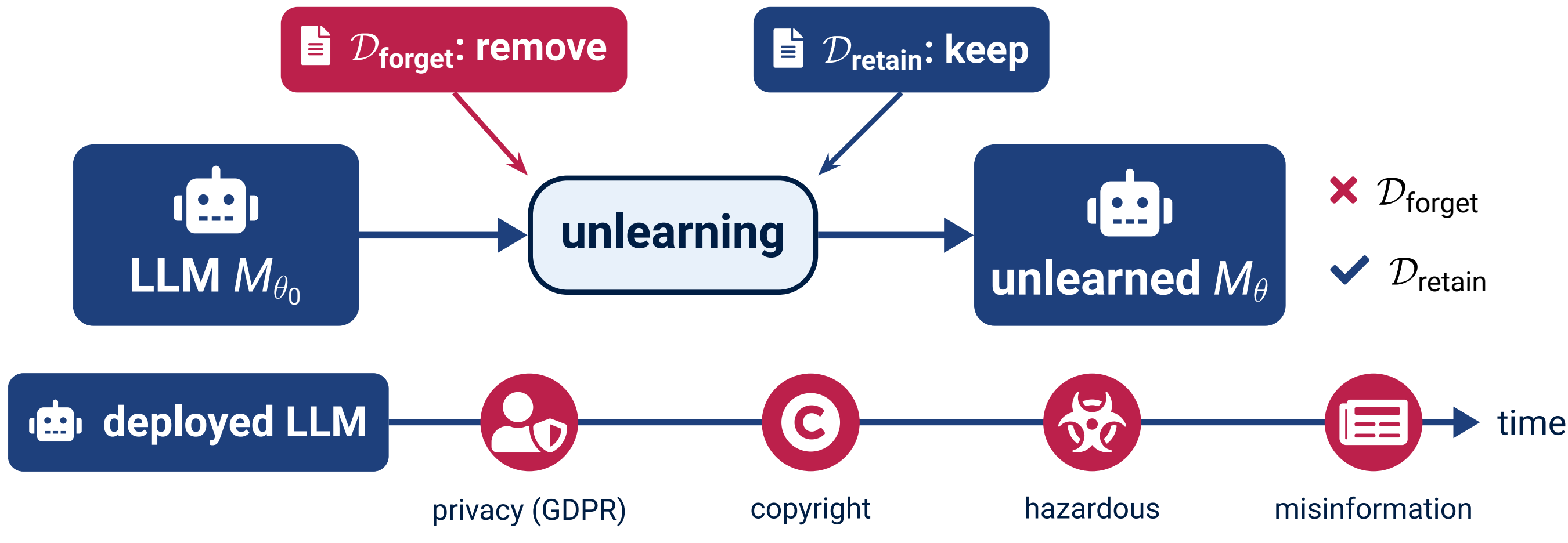
# BLADE: Bilevel Low-rank Augmented-Lagrangian Erasure for LLM Unlearning

Md Toufikuzzaman<sup>1</sup>, Ahmad Mousavi<sup>2</sup>, and Dongwon Lee<sup>1</sup>

<sup>1</sup>The Pennsylvania State University, USA <sup>2</sup>American University, USA  
{mpt5763, dongwon}@psu.edu, mousavi@american.edu

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## Problem Overview



## Objectives

Designing an unlearning method that is ...

|  |                    |                                                      |
|--|--------------------|------------------------------------------------------|
|  | <b>Balanced</b>    | Forgets with minimal model degradation               |
|  | <b>Scalable</b>    | Stays stable as forget set grows                     |
|  | <b>Sustainable</b> | Handles repeated unlearning requests                 |
|  | <b>Robust</b>      | Resists extraction, jailbreak and membership attacks |
|  | <b>Efficient</b>   | Inexpensive finetuning with smaller memory footprint |

## BLADE: Method

### Formulation

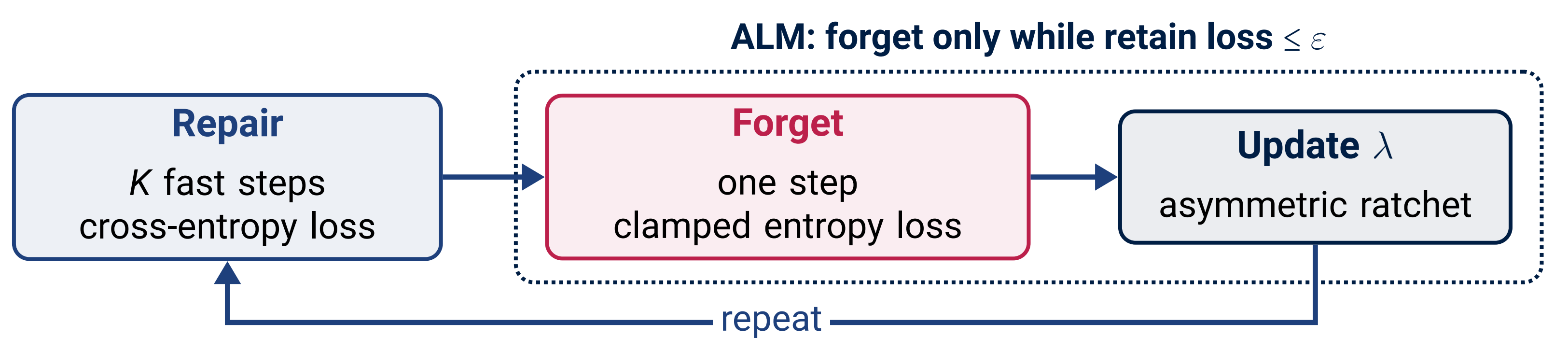
$$\min_{\theta} \mathcal{L}_{\text{fgt}}(\theta^*; \mathcal{D}_{\text{forget}}) \quad \text{s.t.} \quad \mathcal{L}_{\text{CE}}(\theta^*; \mathcal{D}_{\text{retain}}) \leq \varepsilon$$

**Forget** **Retain budget**

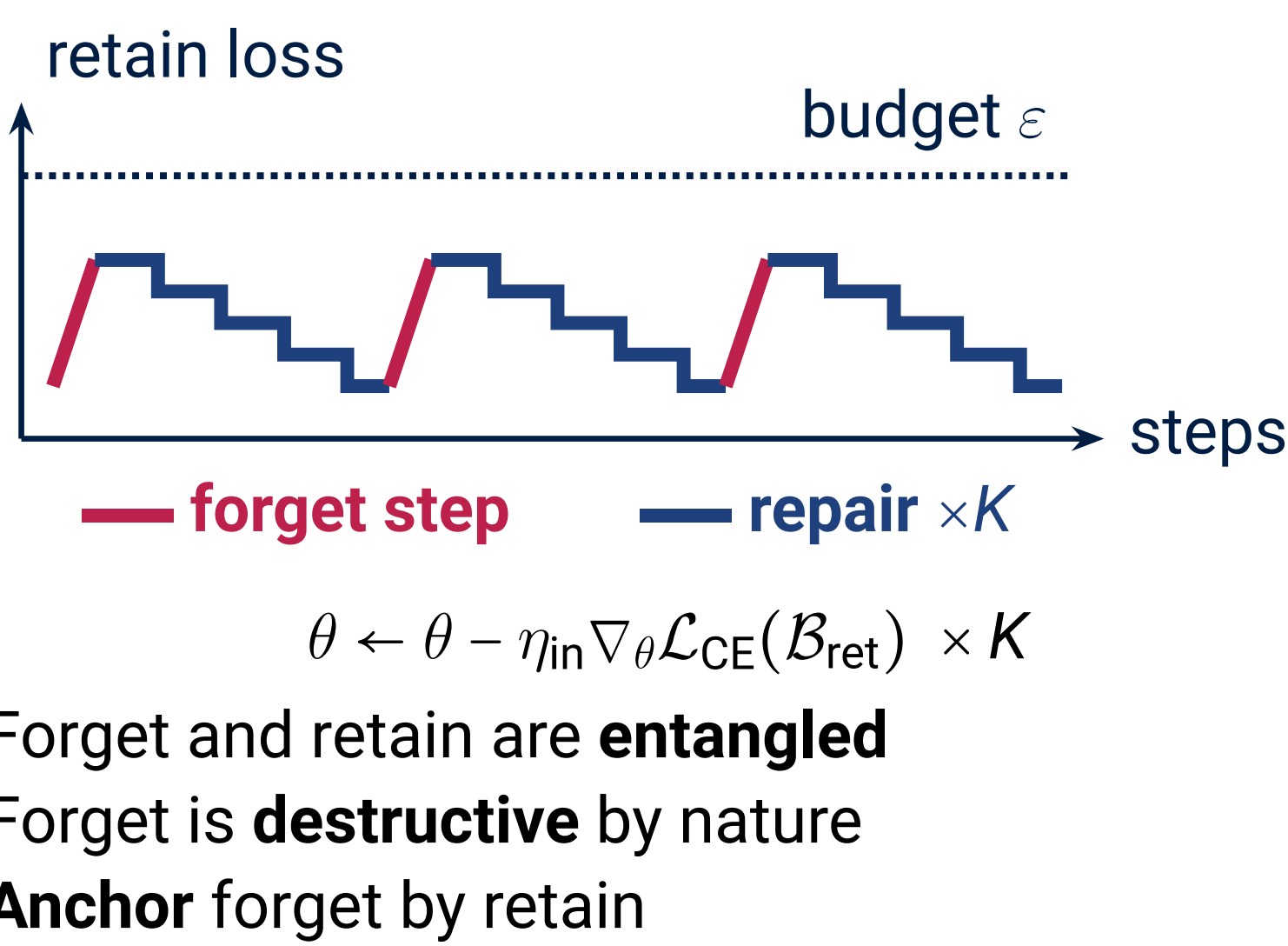
$$\theta^* \approx \arg \min_{\theta} \mathcal{L}_{\text{CE}}(\theta; \mathcal{D}_{\text{retain}}) \quad W' = W_0 + \frac{\alpha}{r} BA$$

**Repair (inner)** **LoRA,  $W_0$  frozen**

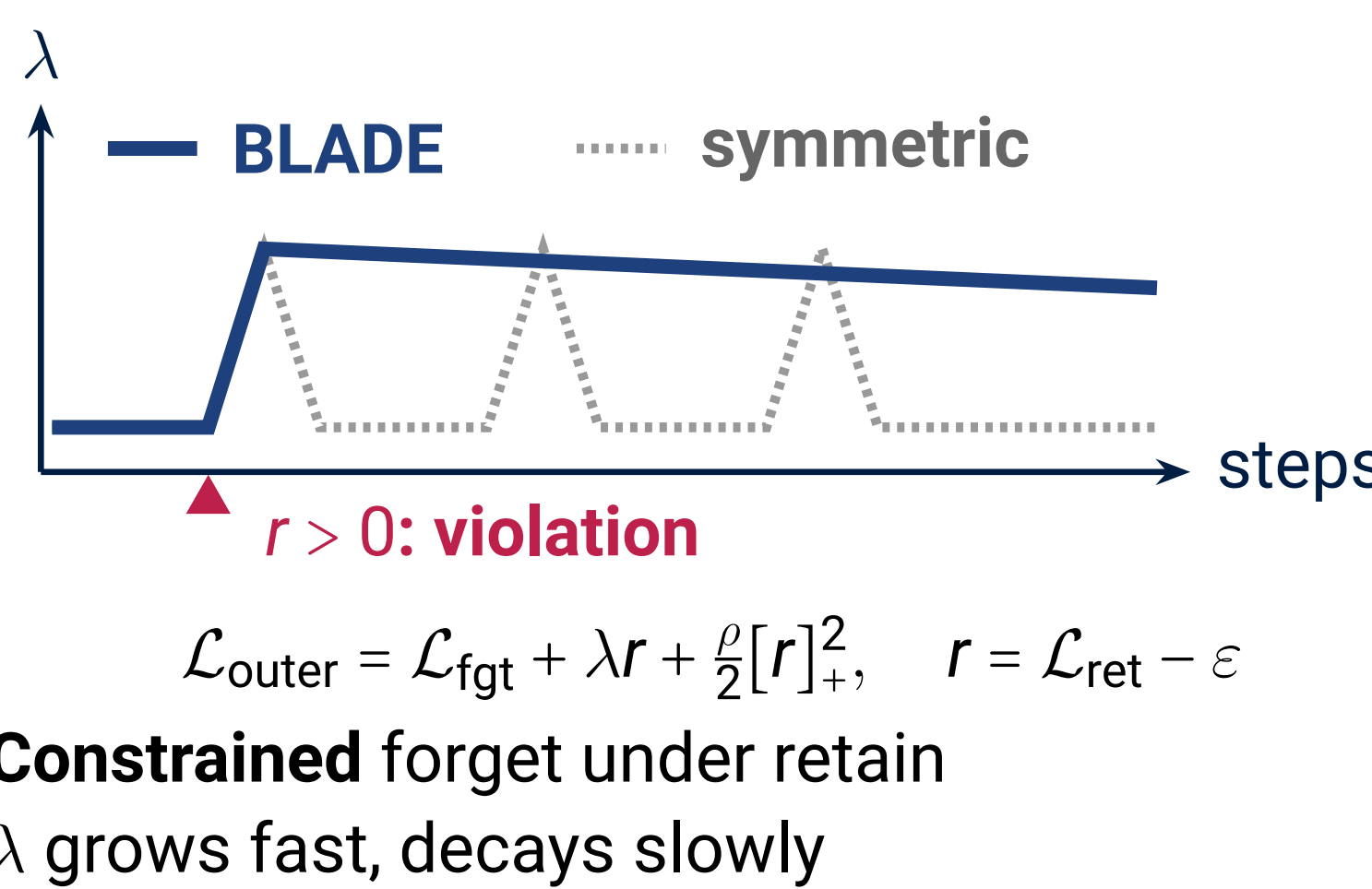
### One training step



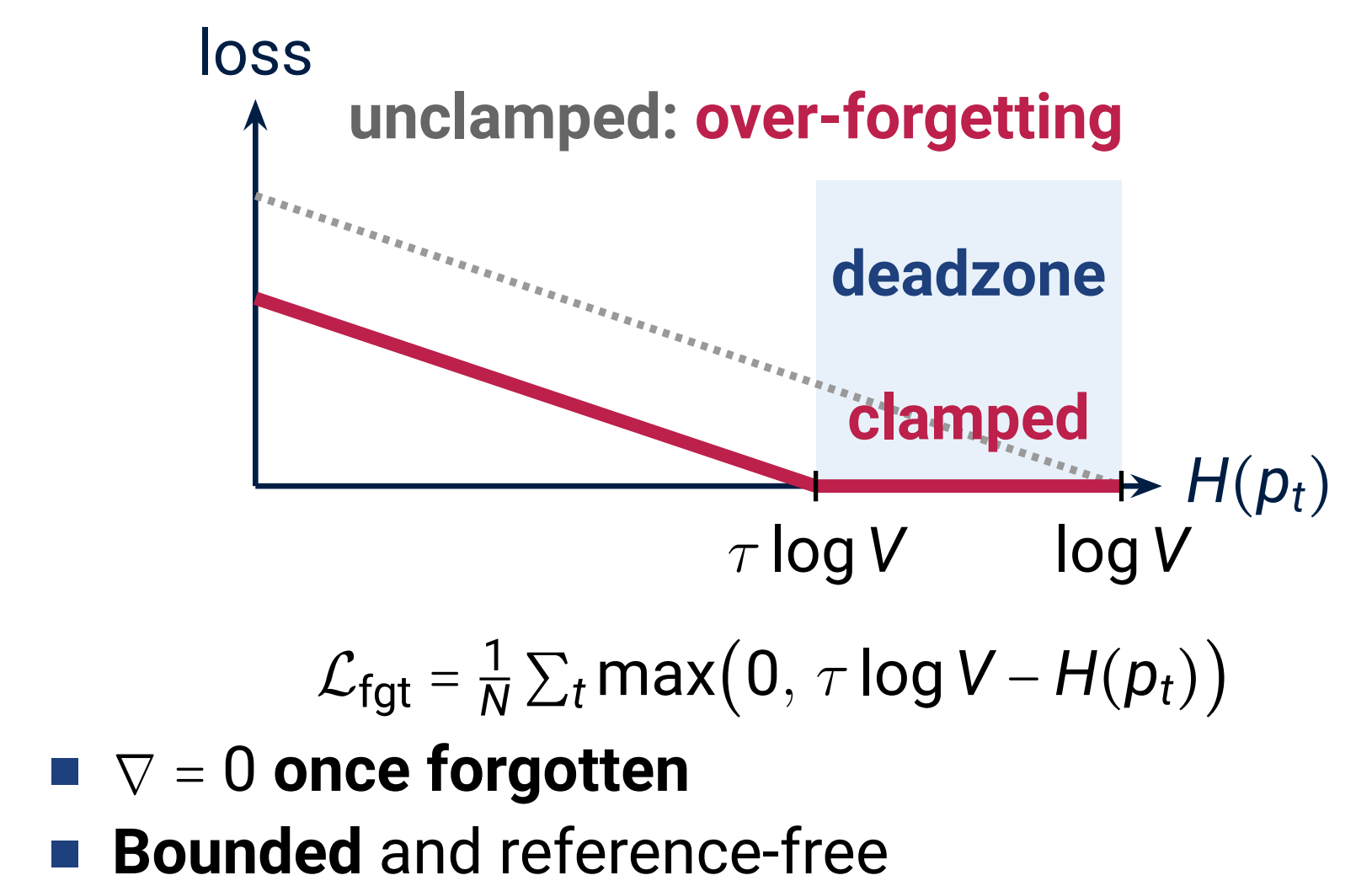
### Repair-first principle



### Asymmetric ALM

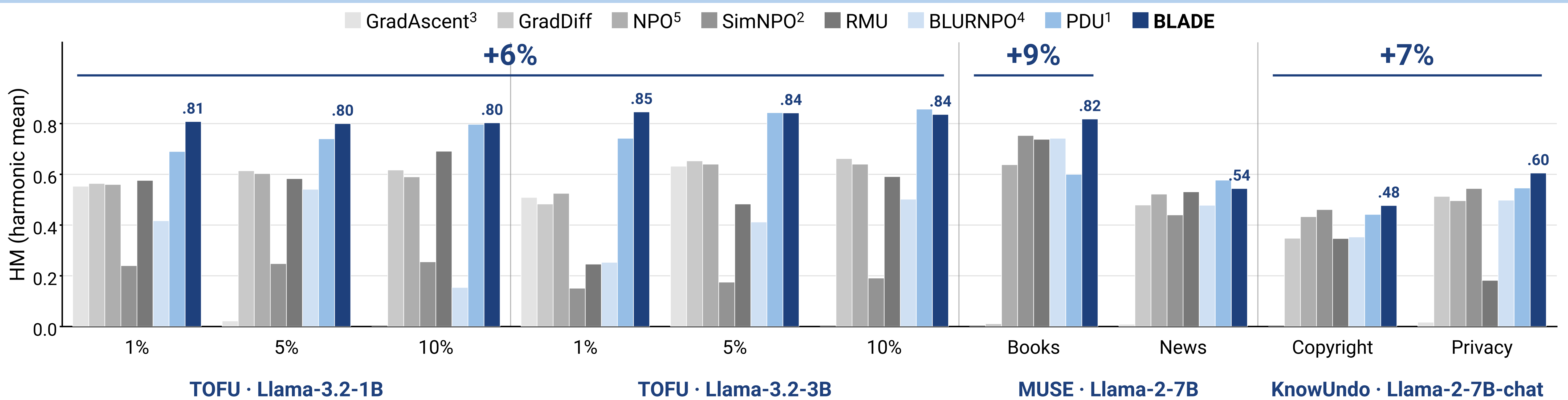


### Clamped entropy

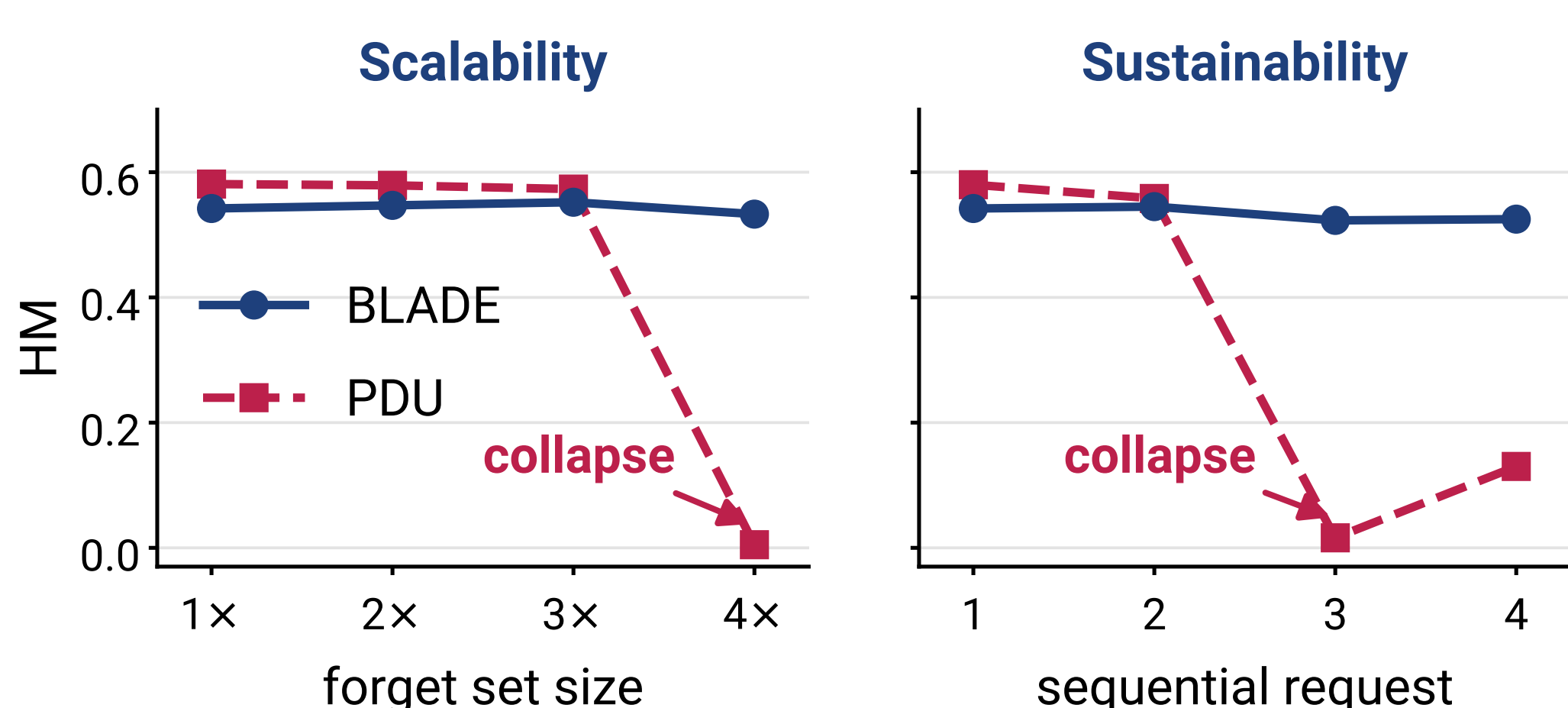


## Experiments

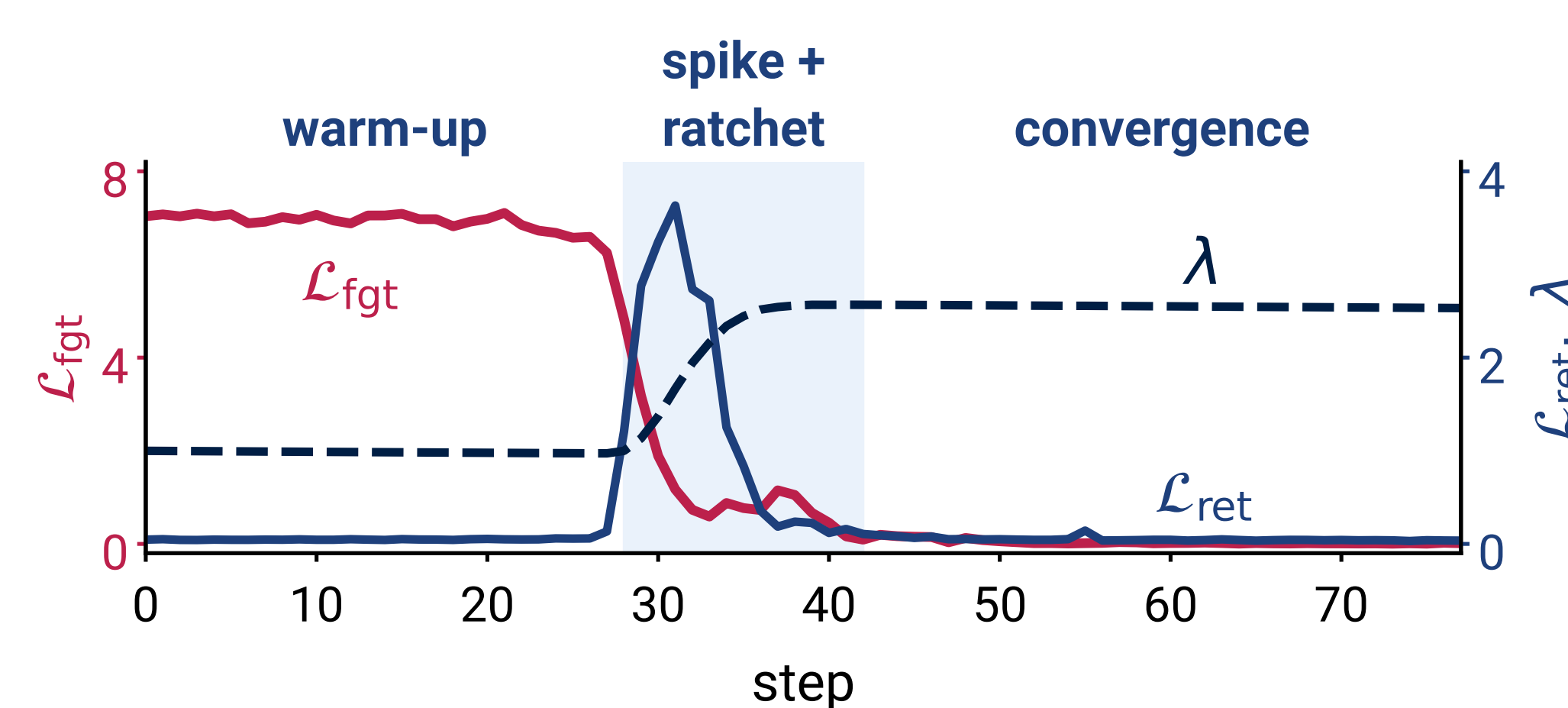
**Dominates baselines across three benchmarks** (avg. harmonic mean of forget and retain scores across 5 seeds)



### Stable under scaling and repeated requests



### Training dynamics of BLADE (MUSE Books)



### Robust and efficient

- Extraction:** best HM
- Jailbreak:** lowest ASR
- MIA:** no leakage
- GCG:** lowest ASR
- 2-5x less GPU memory**

## References

- T. Entesari et al., "Constrained Entropic Unlearning: A Primal-Dual Framework for Large Language Models", 2025.
- C. Fan et al., "Simplicity Prevails: Rethinking Negative Preference Optimization for LLM Unlearning", 2025.
- J. Jang et al., "Knowledge Unlearning for Mitigating Privacy Risks in Language Models", 2023.
- H. Reisizadeh et al., "BLUR: A Bi-Level Optimization Approach for LLM Unlearning", 2026.
- R. Zhang et al., "Negative Preference Optimization: From Catastrophic Collapse to Effective Unlearning", 2024.

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## Paper



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